

 WILEY Trading

THE RISK OF TRADING

MASTERING THE MOST IMPORTANT
ELEMENT IN FINANCIAL SPECULATION



MICHAEL TOMA, CRM

Contents

[Cover](#)

[Series](#)

[Title Page](#)

[Copyright](#)

[Preface](#)

[WHY USE A DATA-DRIVEN APPROACH?](#)

[THERE WILL ALWAYS BE RISK](#)

[A BUILDING-BLOCK APPROACH](#)

[Acknowledgments](#)

[About the Author](#)

[Part I: Principles of Risk Management](#)

[Chapter 1: Foundations of Risk
Management](#)

[A BRIEF HISTORY OF THE LEGISLATION](#)

[DEFINING RISK AND RISK MANAGEMENT](#)

[CLASSES OF RISK](#)

SUMMARY

Chapter 2: Five Steps in the Risk Management Process

OVERVIEW

STEP ONE: RISK IDENTIFICATION

STEP TWO: RISK ASSESSMENT

STEP THREE: RISK CONTROL

STEP FOUR: MEASURING RISK

STEP FIVE: MONITORING YOUR RISK PROGRAM

SUMMARY

Part II: Managing and Measuring Risk

Chapter 3: Predictive Analytics Using Quantitative Analysis

WHAT IS QUANTITATIVE ANALYSIS?

THE DATA-DRIVEN RISK MANAGER

Chapter 4: Statistical Edge and Its Impact on Risk

FINDING EDGE

REDUCING VARIABILITY IN EMPIRICAL PROBABILITY

CRITERIA FOR DETERMINING EDGE

OBTAINING A VALID SAMPLE SIZE

PROBABLE EDGE VS. POSSIBLE EDGE

CHALLENGES WITH CONFIRMATION BIAS
INDICATORS THAT DETECT EDGE
LEVERAGING EDGE USING CONFLUENCE
SUMMARY

Chapter 5: Embracing a Culture of Analytics

RISK MANAGEMENT RULES FOR THE TRADER
THE DATA COLLECTION PROCESS
DATA ASSESSMENT
BASIC PRINCIPLES OF MEASURING DATA
PERFORMANCE MANAGEMENT: WHAT IS THE DATA TELLING ME?
DEVELOPING A TRADE JOURNAL
MEASURING SUCCESS THROUGH A KPI
TRADE REVIEW AND REPORTING ETHICS
BENCHMARKING

Part III: Qualitative Elements of Risk

Chapter 6: The Human Element: Psychological Risks of Trading

LOSS AVERSION
THE RISK OF REGRET
INTUITION TRADING
THE SUCCESS FORMULA AND DISCIPLINE
RISK ACCEPTANCE
MASTERING INCREASES IN SHARE SIZE
THE GRADUATION PLAN

SUMMARY

Chapter 7: Preparing for Risk and Loss

MASTERING RISK DURING A TRADE

THE NEED FOR PLAN COMPLIANCE

RISK PARTICIPATION AND LOSS ACCEPTANCE

PROCESS OF PREDEFINING RISK

TRADER EXPECTATIONS

BUILDING YOUR RISK TOLERANCE

POSSIBLE LOSS VS. PROBABLE LOSS

KNOWING YOUR RISK APPETITE

SUMMARY

Chapter 8: Business Risk Management for Traders

THE FIVE STEPS OF CRISIS MANAGEMENT

SCENARIO PLANNING AND TESTING

RESTORATION REQUIREMENTS

BUSINESS IMPACT ANALYSIS

VENDOR RISKS

PARTNERSHIP RISKS

SUPPORT RISKS

PERSONAL RISKS

CREATING A DEVELOPMENT PLAN

SUMMARY

Bibliography

Index

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The Risk of Trading

*Mastering the Most Important
Element in Financial Speculation*

MICHAEL TOMA, CRM



WILEY

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Preface

From corporate offices to trading desk, the two words “risk management” together have long sparked curiosity, bewilderment, and even confusion. But recently, the profession has received more respect and recognition in our everyday business world and includes a much bigger scope than “those people in legal” or “that guy in charge of insurance policies.”

The Risk of Trading takes a holistic, or enterprise-based, approach that encompasses not only risks on a trade, but also for the trader and ultimately the business of trading.

WHY USE A DATA-DRIVEN APPROACH?

Anyone in the trading jungle knows that a comparison is often made between trading and gambling. This book looks deeply into this analogy and discovers the statistical principles are, in fact, quite similar. When I first mention a risk-based approach to stock, futures, or options trading to the casual investor, the words “risky” and “gambling” often follow. Using a mechanical method to trading supported by a statistically strict data-driven approach often invites even more questions.

“You really don't care where the market is going?” “Don't you want to beat the market?” “Isn't that a bit boring?” “Is that what you do all day?” My response to these alleged market-predictors is, “I don't find winning boring at all. I execute my plan when I have an edge. All the rules are in my plan. What I do is execute my plan with precision just like a project manager building a house or a homeowner

mowing the lawn. I'm not concerned about beating the market. I just focus on executing winning trades, and I only execute when I have a better chance of winning than losing."

There is a certain level of pride when so-called traders talk about their success in predicting market movements. Anyone can do it for a time without any charts, data, or trading experience. You have about a coin-flip chance of being right, and most just hype the few successful predictions over and over again. There's no pressure for them, of course, since most of the time there isn't any hard-earned capital at risk. Performing hours of analysis takes much more energy and certainly comes across as less sexy, if in fact data analysis can have such an alluring appeal. The casual investor tends to be resistant to any form of analytics, since the word itself is reflective of lots of work and analysis. When these methods are presented with simplicity, the better the chance that more will want to hear about them. Combining the conversation with success in trading the markets usually results in a captive audience.

Establishing a culture of trading analytics and metrics is about performing statistical analysis to shift the odds where one can be successful. There is a value proposition in using information and events that have occurred in the past to predict the probability of their occurring in the future under similar circumstances. Although sometimes associated with a geek dynamic, a data-driven approach is evident in many successful areas of business. Service industries currently use customer and vendor data to determine methods to improve quality of service. Internal risk control data generated from process improvement assessments can assist in improving product quality and efficiency. Data-driven decision making is even used to determine if a football quarterback should pass or hand off during certain situations. The book *Bringing Down the House* describes the

MIT blackjack team's winning formula for beating the casinos using data-driven analytics. The book takes a microscopic approach to using data to manage risk and identify those opportunities that present an advantage. Simply put, past data and statistics that have provided positive results in your trading will tend to continue to do so in similar market conditions, thus providing you with the competitive advantage eagerly sought in trading. Combine this methodology with solid risk planning, and you now have access to one of the most powerful trading tools.

Mutual fund companies, hedge funds, and proprietary firms have used risk protocols for some time, but only recently has the average swing or day trader making a living from his or her basement computer had a risk component to trading. It is now commonplace to have a risk section appropriately titled in a trader's trading plan; or at least for those who in fact have created one.

The title of this book reflects the essential principles of the risk management concept required in trading stock, ETFs, futures, or other securities and derivatives. Any type of loss, financial, physical, and even loss of opportunity is a fact of life. If they were not, how we optimize the best use of the ways to minimize loss would be meaningless and unnecessary. One trader's work venue may be vastly different than another's in the same profession. One can perform his or her craft at a hedge fund, prop desk, or at the kitchen table. In this book, I take traders on an assessment journey of their trading business, regardless of venue. The principles of risk theory and the management and application of such ideologies are universal to all who place buy and sell orders for a living.

My personal objective is to build and transform your trading venture into a culture of analytics; simply put, using past performance and historical data to allow you to make decisions that over time will lead you to meeting your

trading goals and objectives. While traditional risk management focuses on pure loss, enterprise risk management, or ERM, widens the scope to incorporate how a trader maximizes opportunity on the reward side. Your historical data are the key to gaining that edge required to be successful.

I often say, if you take away all my knowledge about the markets, I could probably be back on my feet in less than six months. Take away my data and I'm out of the business. For those concerned by being bombarded with information on variance, probability, statistical theory, and central tendency theorems, I understand your pain. Risk management conferences are filled with reinforcement of the driest of educational topics. Rather than put you through the pain and perhaps boredom of educating you on the latest and the greatest holy grail, these essential analytic principles are explained in a direct no-nonsense manner using examples that you can immediately use in your daily trading.

My first book, *Trading with Confluence*, focuses on data analytics where clusters of different data conclusions all lead to the same action. *The Risk of Trading* captures the essence of my first book and expands the confluence trading concept using data mining and trade data analysis. All this fun with data interwoven with the risk management process somehow has to be summarized in a format that can allow for new decisions to take root.

The objective of an enterprise risk management platform is to act as a repository for all direct and indirect exposures to losses associated with trading. While most focus on trade management and loss of a trade, a true risk manager is aware of all potential risks to his or her trading business, understands the potential impact of each, and has mitigation plans for those worthy of concern.

THERE WILL ALWAYS BE RISK

One of the most popular questions I receive is related to the relationship between risk management and stop-loss orders on a trade. The chapter on risk control deeply penetrates the stop-loss technique as a valid tool in one's risk management arsenal. My biggest concern, however, is the notion that the scope of risk management that protects one's trading account and livelihood, sadly, lies in the use of a simple bracket order feature provided free on a brokerage trading platform. My personal goal with this text is to widen the scope of this topic for traders who believe managing trade risk simply means using stops.

As long as we perform an activity, no matter how much we try to eliminate loss from anything we do, we face the fact that some form of loss could occur. Exposure can strike anyone or any business at any time. Businesses seeking a profit or a charitable organization seeking to provide services with a challenging budget both perform activities that are subject to loss. We perform activities that are subject to a form of loss every day in our lives, and it can interfere with our ability to achieve our desired objectives. As traders, we have ventured into a business where the chance of loss is one of the basic tenets of speculative activity, yet many ultimately find it impossible to accept.

Despite the vast opportunities and possibilities of loss that can occur, the risk management process exists to minimize the frequency and severity of such disruptions in a quest to reach our trading objectives. Being able to apply this process to your trading business is ultimately the goal of this text. Taking a risk management approach and applying the risk principles to trading decisions will allow you to effectively manage your trades and your business.

A BUILDING-BLOCK APPROACH

The goal of this book is to provide you with the fundamental and applicable knowledge required to implement risk management principles in your trading. Ultimately the objective is to see vast improvement in trading performance and management of your operation. In order to implement and apply a solid risk-based strategy, it is essential to have an advanced understanding of risk theory and to continuously apply the steps within the risk management process. The book is therefore structured in three parts that build upon one another.

Part I acts as a foundation course for the reader to understand how we define risk and to understand the risk management process. The book discusses this five-step process extensively, because it is essential to have a thorough understanding of the decision-making process before attempting to apply the principles to your trading. In addition, enterprise risk management models will also be introduced as an effective methodology to protect trade management as well as the entire pool of risks that can affect your trading business. Whether you are a sole-entity trader trying to make it big or a structured firm seeking to optimize operational efficiency, the use of ERM models in your trading plan can improve your key performance results in a relatively short period of time.

Once you have obtained a solid understanding of the risk management process, Part II advances toward the level of utilizing practical applications to manage and measure risk using data analytics. Here we go inside the mind of a risk manager as he or she tackles the upcoming trading day and learn how to leverage solid trading strategies within a culture of metrics. Chapter allows the reader to learn specific strategies that provide an advantage and to execute them within a variety of applicable risk-to-reward scenarios. You have undoubtedly heard that you must keep

track of your trades in a trade journal. Chapter discusses what data to capture and how to convert your journal data into a profitable trading machine using key performance indicators. I'll take you by the hand during the reporting process to determine how best to marry results with key performance measures and benchmark them to a common standard. If you have experience in the information technology world, you may find the teachings on business intelligence quite familiar. Here we introduce our ERM framework using data analytics and allow it to breed new life within the vast assortment of business intelligence software once only available to large hedge funds.

Part III has a psychological overtone to it and discusses the human risk elements that challenge every trader. No matter how many trading plans or trade journals I have assessed in my work, greater than 90 percent of all risk deficiencies are connected to a psychological root cause. Comments such as "I changed my plan because ..." or "I just felt that ..." and "I didn't want to take the loss so I ..." are quite popular comments in trade journals. I easily could have titled this section "Reality" due to the bold instruction of some of the not-so pleasant parts of the trading business. Managing losing streaks and drawdowns is one of many psychological topics covered. Chapter discusses how to manage your psychological risk capital and teaches how to use the appropriate share size for each trade. The "graduation plan" structure takes away the emotional decision making regarding size and effectively allows you to build your trading empire. The effective risk control and developmental techniques discussed can be applied immediately after reading this book.

Chapter provides the information on how to prepare for loss and how to position trade risk and reward models most effectively, allowing you to capture the most opportunity from each trade. Chapter dives deeper into the most

overlooked risks that affect nearly every trader at some point during his or her career. Computer systems go down, data get corrupted, organizational structures change, personal conflicts arise, and sometimes traders gets a call from the school nurse saying little Britney caught a cold 10 minutes before a major Fed decision. They all involve decisions that need to be implemented to minimize loss, and we review how personal conflicts can impact your trading. Also discussed is how one assesses business continuity risks and how to minimize critical and often costly downtime as a trader. Understanding brokerage firm risks and the use of other vendors is also brought into the light in understanding their role in your day-to-day business planning, a must-read for those serious about becoming successful over the long-term.

You are on your way to exploring the avenues that guide you to being a more consistent and profitable trader. Whether you are trading as an individual or part of a trading firm, incorporating risk principles into your trading will strengthen the inner core of your venture and potentially provide that missing link for those who have the hunger but need more to succeed. I wish you all the best in your risk journey ... and it begins now.

Michael Toma, CRM
November 2011

Acknowledgments

One would think that a second book on trading would sail smoothly from draft to bookshelf. I've learned a lot about the writing and publishing business by taking the initial concepts and bringing them to life in print. This was not the case, however, with *The Risk of Trading*. This book is designed for a broad audience of new traders to the established trading firms. There was also a need to consider the international audience, who prefers a different writing style and content than their Western counterparts. In the end, the project was completed with the help and inspiration of many. Many thanks to the team at John Wiley & Sons for their expertise, commitment, creativity, and ability to continuously keep me razor focused on allowing the vision to become a reality. I am so honored to be a part of the Wiley family.

In the trading world, I am extremely grateful for the opportunity to work with the professionals at MoneyShow.com. Any individual with a different concept needs an audience to be heard. These online and tradeshow pros really know how to provide the best forum for an average person from New Jersey to have the opportunity to give back to the trading community; they provided me with the educational foundation to become the trader that I am today. The venues also provided me with the opportunity to develop my trader and corporate network that has led to other ventures in the trading and risk management arenas. Whether I'm writing articles, taping educational videos, or presenting at the national expos, I'll always have the team at MoneyShow.com to thank.

The opportunity to branch out into teaching risk management and trading needs more than just a stage. It

requires a person who believes that you not only should be heard but can capture an audience that wants you to be heard. Tim Bourquin at Trader Interviews took a chance on me. The flywheel started turning, and it has never stopped since. From subsequent interviews to video interview segments at the expos, Tim has been not only the quintessential business leader but an ethical professional who was willing to give an unknown a chance well before becoming one of the “young guns” on the expo circuit. I promise I will pay it forward.

I've crossed paths with many traders in my career, but one that has stood out as a great colleague and teacher is market profile expert Rick Vinecki. He is a master of market profile, and the trading concept has allowed many a paid vacation ever since. He is the ultimate pro in a business that needs more of his ethical professionalism and trading talents he brings to his students each trading day.

All the paths that have led me to this point would not have been available without The National Alliance group, which founded and supports the Certified Risk Management program. I am honored to hold such a prestigious designation and am well aware that the opportunities that I have been blessed with would not have been possible without their assistance. I look forward to working with you and continued writing through the Research Academy program for years to come.

In order to fulfill such dreams of being a writer, speaker, and educator, I needed the core professional competencies and mind-set required to allow the ability to succeed. I thank John Oliveira for providing the encouragement to always be thinking outside the box and to expand the limits of my destiny. This wisdom and spirituality has not only allowed me to become a better-rounded professional, but also a better giving person. Thanks to you, my journey is measured only by the number of lives I can enrich.

My family certainly deserves many thanks for their kindness and support during the writing journey. There was always a fresh pot of coffee on whenever I called up and said I needed a break from the book. My parents may not completely understand the concepts of such “a very involved book,” but their telling me they are proud is more than enough fuel to allow me to complete such a challenging project.

Finally, thanks most of all to my darling wife, Nancy. Her encouragement for me to accept such an opportunity during some very long trading weeks will always be appreciated. Her sacrifice of sitting home during some beautiful summer weekends in New Jersey or editing a chapter or two in Shanghai just so I can complete this book will never be forgotten. Having your encouragement makes me believe that I can accomplish anything. Thanks to your love and support, I can truly say I've only just begun.

About the Author

Michael Toma is a corporate risk manager and specialist in trading the equity index and futures markets in the United States. A Certified Risk Manager, Toma is a frequent speaker on the industry trading conference circuit. His first book, released in 2010, titled *Trading with Confluence* (Outskirts Press), provides a first-hand perspective of risk-based trading and the challenges that new traders are forced to overcome. In addition to managing his own international risk management consulting firm, he is an active member of several risk management organizations and research academies that promote corporate ethics and risk management educational programs.

PART I

Principles of Risk Management

CHAPTER 1

Foundations of Risk Management

The golden age of corporate and accounting scandals during the Enron era placed risk management on the lips of the average corporate employee. In an attempt to mitigate future scandals, the U.S. government passed the Sarbanes-Oxley Act in 2002. The U.S. Senate called it “The Public Company Accounting Reform Act,” and the House called it the “Corporate and Auditing Accountability and Responsibility Act.” I'll refer to it by its simple hybrid name, SOX. This legislation is at the heart of risk management today, so it's vital to begin any discussion with a little history. The goal of Chapter 1 is to provide an introduction to risk principles, which you will use in your day-to-day trading activities.

A BRIEF HISTORY OF THE LEGISLATION

The SOX Act brought new regulation to corporate board member responsibilities, oversight of accounting practices, corporate governance, fraud, internal controls, and enhanced financial disclosure. While there are debates over the effectiveness of the actual bills, mostly from cash-strapped companies that find the cost of SOX compliance to be more expensive than paying the fines, the need for managing risk has bubbled to the surface as one of the

must-do's on the corporate budget sheets. Many companies that are not required to comply with SOX attempt to be within compliance due to the best-practice models inherent in the act's bylaws. Other risk control standards, such as the SSAE 16 assessment championed by the CPA community, also adjusted their prize audit standards in an effort to put on a cleaner pair of SOX.

Since the meat of the act involved the proactive identification, assessment, control, financing, and monitoring of risks, what better individual to implement such a monster than your risk management professional. Since 2002, a more holistic form of risk management known as enterprise risk management (ERM) has grown in popularity. Rather than just identify problems, the ERM specialist can also be viewed as a process improvement and efficiency resource that can assist in improving the bottom line. Technology, database management, data analytics, and the use of data-driven decision making has also resulted in a greater acceptance of the ERM framework. Along with a never-ending surge in data technologies, business analytics, and the sudden urge to measure everything that's measurable, the two have brought risk management to the forefront of today's trading. Many of these ERM models continue to grow in popularity and have a nouveau-risk appeal. Quickly fading are the more traditional risk models that have stuffed the file cabinets of institutions that see risk management merely as insurance policy management.

Using math and numbers to make proper trading decisions should be simple and transparent in communicating the value proposition for using such methods. There has been a dramatic shift in focus over the most recent decades regarding the application of financial risk management. It is important to be clear that the risk principles and theories noted earlier are a critical requirement in your ability to be successful. During the same time, data-driven decision

making has accelerated the thirst to incorporate risk management into trading decisions and now is a critical component of any institutional process. In other words, “what is the data telling us?” is equally as important in directing managers on what to do as relying solely on theoretical and strategic methodologies. Many of the texts focus on trading theory only, and I find myself being extremely critical of this information when used in a live environment. The common securities industry phrase “past performance is not indicative of future results” may have some merit (particularly from a firm's legal perspective), but data analytics is founded on the principle that patterns or behaviors tend to repeat when similar market elements are present.

The term *risk management* has been used in an ever-increasing fashion in recent years. Regulatory changes, the need for greater compliance-based operations, the increasing cost of loss, and evidence of its impact on the bottom line has fueled discussion points on the previously closed-door topic. This chapter introduces the elements of risk management and the steps used in the process. We will also review the benefits of infusing risk concepts into your trading plans and operation.

It's important to note that people see risk differently in their professional roles. Of course, we focus our discussion on risk within the scope of a trading environment. Those in the mortgage industry, particularly underwriters, may see risk as the potential default on a loan. Mortgage brokers may see an increase in the fixed rate as a risk to their business, knowing that an increase in mortgage rates would slow demand. At the same time, other brokers may like a spike in rates that create a demand to lock in low rates or opportunities to convert those with adjustable-rate mortgages to a fixed-rate plan. Still others in the industry may see the regulatory agencies as their biggest risk since

the passing of stricter processes or disclosure may hinder their ability to generate revenue from specific market segments. This was exemplified during the aftermath of the 2008 mortgage crisis when mortgage brokers were required to stick to tighter regulatory processes as part of the lengthy Dodd-Frank bill. In a hospital operation, a surgeon may see the primary risk of a procedure not working or even the death of a patient, while upstairs in the administration department, the risk of rising medical equipment costs may be on the identified list. Down the corridor, the legal department's top monitored risk is the public relations plan gaps that may be exposed during a crisis. As we progress through the risk management process, it is important for readers to establish their operational trading goals before actually identifying potential trading risks. Let us first define risk, at least within the context of a trader, before we proceed to determine how to manage it.

DEFINING RISK AND RISK MANAGEMENT

Various definitions of risk management are used depending on the industry, source, and purpose. For example, corporations tend to use a definition to define the purpose of their department with the same name. Others use their department's mission statement as the foundation for how the term is defined. The basic theme for most definitions of risk management includes the decisions made to protect the entity's assets and minimize the adverse effects of activities we perform in order to attain company goals. In the trading world, the term *assets* can refer to many things. Logically, our trading capital is probably one of our most essential assets. Capital in trading is the ticket to the show. Simply put, if there is no capital, there is no business; hence the important link of risk management to capital. Later we will

discuss the risk identification process in depth. It is critical to also define the logical classification of a trader's capital or related company assets before the risk management process takes place. If not, how can we protect an asset that we haven't identified as an asset?

If risk management is based on the premise of protecting assets, we need to understand the term *risk*, bypass the lip service often given to the term, and define it, at least within the scope of this text. Being a professional risk manager and trader for many years, I've come across several definitions of this simple yet complex word and would probably receive 10 different definitions of the word if I asked as many people. The best definition that reflects the purpose of pure risk management and maintains its simplicity for explanation purposes is simply the chance or uncertainty of loss.

Pure vs. Speculative Risk

The trading community is certainly aware of the chance of loss on any given trade or trading strategy. We accept risk not so we have the opportunity to engage in risky activity (although our business is a magnet for those who relish risk taking), but to reap the opportunity of rewards or capital gain. The scope of pure risk is limited to those activities and decisions undertaken to avoid loss without the rewards found in speculative risk taking.

The insurance industry is based on an individual or entity's need to reduce loss without any opportunity to benefit from an event or decision. The term used in insurance is *indemnification*, or "to make whole." If you review homeowner, auto, or similar policies, you may not have to search deep to find the words indemnity or indemnification. Many have it as the first word on the policy right on top of page one. The basis of indemnification is to bring the owner of the policy's asset or use standard to where it had been

before the loss. A car gets repaired, a home gets rebuilt, or stolen property gets replaced. Even in life insurance, the value of the policy is considered indemnification to the beneficiary, at least for policy definitions, upon the demise of an individual.

Speculative risks include a chance of indemnification plus additional gain. The desire for this gain is offset by the potential loss that may occur. The gaming industry is a perfect example of speculative risk at work. Companies allow customers to partake in speculative ventures, described in their eyes as entertainment. Casinos and financial markets provide the forum for speculators to execute their speculative risk-taking activities. In an ironic twist, both industries would be nonexistent if the speculator did not benefit from time to time. While reducing player odds in a casino would greatly benefit the house in the short term, these companies know that there needs to be the potential for speculation to pay off, whether it be in winnings or entertainment value, to maintain solvency. Gaming companies are experts in statistical probability, valid sample sizing, and mathematical edge, the same formulas used by successful risk-based traders.

As you read through the book, it is important to remember that although traders are generally placed in the box of speculators, they are subject to both pure and speculative risks. We limit our loss on a trade with the intention of making a profit of the same or even greater amount. This is speculative risk in its simplest form. A trading platform crashing in the middle of a trade can only be defined in the pure risk category. It is this lack of inclusion of both pure and speculative risk types in the risk management process that is one of the biggest identified concerns in my auditing work.

Scope of Risks

When performing risk assessments in the trading community, I frequently widen the scope of the pure and speculative risk definitions noted above. It's normal for traders to be concerned about their performance and desire to continuously improve. After all, that is what puts bread on the table. In determining chance of loss or level of uncertainty, always incorporate an element of variance into the definition of risk. In other words, risk is also the possibility that any outcome may be different from an expected outcome. For example, a trader may be extremely satisfied with reaching his or her profit goals for a defined period. A risk manager will assess the variability in outcomes that had resulted in the profit goal. Perhaps the trader took on too much risk or violated trading rules to obtain the goal. Maybe the goals are too conservative for the trading instrument or share size is too large, thus allowing the trader to easily reach the goal and allow psychological comfort for that period. As we walk through the risk management process, please be sure to consider the possibility of a variation of outcomes that may differ from the intended result.

CLASSES OF RISK

Although we will devote intensive study to risks directly impacting traders, such as trade risk or the potential for loss on a trade, it is common to find other classes of risks that typically impact traders' performance or overall business at least once during their careers. Here are some classes of risks that tend to be overlooked by the typical trader.

Industry Risks

Exposure to trends in the financial marketplace or financial structure often can impact traders' ability to meet their goals. Changes to instrument liquidity and bid/ask spreads

can make a dramatic change to a trader's bottom line. Competitive product releases, particularly in the exchange-traded funds (ETF) community in recent years, while allowing a supermarket of trading choices, have also have resulted in increased liquidity of some products. Traders who claim to be experts in one product expose themselves to a learning curve risk when forced to learn to trade new products.

Political, Legislative, and Juridical Risks

Legal changes and government interpretation of policies and laws can have an impact on the ability to trade and even directly to a trader's bottom line. A “trader tax” has been in discussion for many years now and has increased in the spotlight since the economic crisis of 2008 and the emergence of national debt reduction ideas floating around in the legislature. One proposal called for a small tax on each trading transaction. Although it would be just a few pennies per trade, many traders who tend to scalp for pennies or ticks at a time could be exposed to a US\$25,000+ impact. The “Let Wall Street Pay for the Wall St. Bailout Act” surely would be enough for some smaller accounts to close their trading doors. As a trading professional, it is always prudent to keep abreast of the latest discussions in the political arena that can impact your ability to trade and the potential effect on your bottom line. A good risk management plan would include a diversified strategy to seamlessly transition from one instrument such as stocks to futures or forex or vice versa should political or regulatory risks emerge. The structural transition may only take one or two days in the form of opening a separate account or adding the trading feature to an existing platform. The true risk exists in the variation of results that may occur until reaching the level of trading precision with the new instrument.